

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **data pipelines** using **Python**, **SQL**, and **ETL frameworks** to ingest, transform, and aggregate enterprise data from **PostgreSQL**, **MySQL**, and **Oracle** databases; reduced data processing latency by **40%** through optimized query logic and incremental load strategies.
- Built interactive **Power BI dashboards** and **Tableau visualizations** for executive stakeholders in payments and financial services, modeling complex datasets with **DAX measures** and **Power Query** transformations; enabled real-time operational visibility and supported **SLA tracking** for **P1/P2 incident response**.
- Developed machine learning models using **scikit-learn**, **pandas**, and **NumPy** for predictive analytics and anomaly detection on production transaction data; applied **statistical methods**, **hypothesis testing**, and **A/B testing** frameworks to validate model performance and business impact.
- Orchestrated end-to-end **data engineering automation workflows** using **GitHub Actions**, **Jenkins CI/CD pipelines**, and cloud orchestration to schedule and monitor batch jobs; integrated **AWS Lambda** for serverless data processing and **CloudWatch** for pipeline observability, achieving **99.5% uptime**.
- Analyzed large-scale enterprise data using **big-data concepts** including **Spark SQL** and **Hadoop** ecosystems; generated statistical insights and time-series forecasting on payment message volumes and system performance metrics for capacity planning and regulatory reporting.
- Led cross-functional data requirements gathering with banking-domain business analysts and operations teams; translated complex business logic into reproducible data models, implementing ISO 20022 and SEPA payment reconciliation workflows; documented data lineage and transformation rules to ensure audit trail compliance.

SKILLS

Data Science & Machine Learning – scikit-learn, supervised learning, unsupervised learning, predictive analytics, statistical analysis, regression, hypothesis testing, A/B testing, evaluation metrics, model validation.

Data Engineering & ETL – Python, SQL, data pipelines, ETL frameworks, data warehousing, Apache Spark, Hadoop, batch processing, incremental load strategies, data transformation.

Business Intelligence & Visualization – Power BI, Tableau, DAX, Power Query, data modelling, interactive dashboards, business reporting, operational analytics.

Programming & Cloud Technologies – Python, Java, SQL, REST APIs, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, GitHub Actions.

Databases & Data Platforms – PostgreSQL, MySQL, Oracle, database optimization, query performance tuning, schema design, data validation.

Domain Expertise – financial services, payments processing, ISO 20022 message flows, SEPA, banking operations, production support, SLA management, incident response, GDPR compliance.

Tools & Frameworks – Jenkins, Jira, Confluence, RStudio, R, pandas, NumPy, Flask, FastAPI.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

SWIFT Payments Operations Dashboard & Data Analytics (HCL Technologies)

- Designed and maintained a production-grade **data warehouse** ingesting **SWIFT messaging** data (**MT/MX formats**) and ISO 20022 payment flows; modeled complex transaction hierarchies and reconciliation workflows using **PostgreSQL** and **SQL** schema design to support **SEPA and cross-border payment** analytics.
- Built operational **Power BI dashboards** with **DAX measures** and **real-time data refresh** to track payment message volumes, processing latency, and error rates; integrated **CloudWatch metrics** for system health monitoring, enabling **SLA compliance tracking** and **P1/P2 incident response** decision-making.
- Developed **Python ETL scripts** and **SQL stored procedures** to extract, validate, and transform transaction data from interconnected ERP and payments systems; automated daily reconciliation reporting with **audit trail logging** for regulatory and GDPR compliance requirements.

Ireland Jobs Tracker – AI-Powered Data Pipeline & Automation (Personal Project)

- Engineered a production **data pipeline** orchestrating job description ingestion from **Apify web scrapers**, leveraging **Claude API** and prompt engineering to parse and structure job requirements into **JSON** format; deployed as **serverless functions** with **GitHub Actions CI** for scheduled automated execution.
- Implemented statistical analysis and feature extraction workflows using **Python**, **pandas**, and **NLP techniques** to identify skills, salary ranges, and hiring trends across job listings; published interactive **data visualizations** and **analytics reports** to **Vercel** for stakeholder consumption.
- Designed end-to-end automation architecture with error handling, retry logic, and observability; applied software engineering best practices including version control, CI/CD, and infrastructure-as-code to ensure reliability and reproducibility of data workflows.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)